

Data-Driven and Internal Model Control for Optimal Output Regulation of Linear Systems with an Assured Convergence Rate

Hanwen Cai
Department of Automation
Xiamen University
Xiamen, China
hanwencai@stu.xmu.edu.cn

Weiyao Lan
Department of Automation
Xiamen University
Xiamen, China
wylan@xmu.edu.cn

Xiao Yu*
Institute of Artificial Intelligence
Xiamen University
Xiamen, China
xiaoyu@xmu.edu.cn

Abstract—This paper addresses an optimal output regulation problem with a requirement on convergence rate for linear time-invariant systems with unknown system dynamics. First, an internal model is introduced to transform the problem into a dynamic optimization problem. Then, a new virtual system is designed to replace the original system and internal model. By collecting the data from the virtual system, a novel data-driven learning algorithm is proposed, which not only learns the optimal control policy but also guarantees the convergence rate of the tracking error to be no slower than a predefined convergence rate. Finally, an illustrative example is presented to validate the proposed approach.

Index Terms—Data-driven learning control, optimal output regulation, assured convergence rate

I. INTRODUCTION

The output regulation problem has garnered significant attention in recent decades, aiming at designing a controller for the plant system such that the closed-loop system is able to track desired trajectories and reject external disturbances [1]–[3]. By further minimizing a predefined cost function, the closed-loop system's response achieves optimality in certain aspects. Such a problem related to linear systems is known as the linear optimal output regulation problem, which has been studied in [4]–[6]. However, these methods require known system models, which may not always be feasible in practice.

When the system dynamics is not available, adaptive dynamic programming (ADP) and reinforcement learning (RL) approaches were adopted to tackle the optimal control problem using online system data; see for examples [7], [8]. Specifically, in [9], the authors proposed an online ADP-based learning algorithm to deal with the linear quadratic regulation (LQR) problem. Then, the approach were further developed to address several control problems, including the output regulation problem [10], the LQR problem with output-feedback control [11], and optimal control of continuous-time nonlinear systems [12], etc.. Later on, some RL and ADP algorithms were presented in [13], [14] to address the LOORP with completely unknown system dynamics.

Feedback-feedforward and internal model principle are main ways employed to solve output regulation problem. Feedback-feedforward control transforms the LOORP into an optimal

feedback and an optimal feedforward control problem, which was employed in [10], [15], [16]. However, all these works need to solve both the algebraic Riccati equations (AREs) and the regulator equations. In contrast, internal model principle turns it into an optimization problem and is free from the solutions to the regulator equations, as applied in [14], [17]–[19] for linear continuous-time systems. These two methods for solving the LOORP are realized by policy iteration (PI) or value iteration (VI). PI algorithm usually requires an initially stabilizing policy [20] to ensure its convergence, while the VI algorithm can start with any initial control policy.

In most of the aforementioned works [9], [10], [13], [15], [21], the authors constructed iterative learning equations (ILEs) by a number of finite window integrals (FWIs), which required storing information-rich data points along the system trajectory. To overcome the difficulties, a novel two-layer filter based adaptive optimal control algorithm with initial excitation was proposed in [22]. This approach eliminated the requirement of known state-derived and past data storage. However, the implementation of two-layer filters may lead to an increase in computational load.

The convergence rate of the closed-loop system is also of importance in practical applications. In [23], the finite-time adaptive control was studied in a non-strict feedback form based on the system dynamics. In [15], the learning-based LOORP with convergence rate requirement was investigated. However, the method has its applicability limited to the case in which the system dynamics are partially unknown, and requires solving the regulator equations.

This paper presents a novel data-driven learning algorithm to solve the LOORP with an assured convergence rate. The main contribution is twofold. First, compared to [15] where the system matrices are partially required to solve the LOORP with an assured convergence rate, we solve this problem when the system matrices are completely unknown by using the internal model. Second, in contrast to [17], [24] which required the internal model state for implementing the learning algorithm, our proposed learning algorithm simply requires a user-defined signal with an appropriate bound in the learning process. Moreover, we construct the iterative learning equation without requiring finite window integrals.

The remainder is organized as follows. In Section II, we present the preliminaries and problem formulation of the LOORP with an assured convergence rate. In Section III,

This work was supported in part by the National Science and Technology Major Project under Grant 2021ZD0112600, and in part by the National Natural Science Foundation of China under Grants 62173283 and 62273285. (Corresponding author: Xiao Yu.)

we propose a data-driven learning method and present the solvability analysis. In Section IV, we provide a simulation example to show the validity of the proposed approach. Finally, conclusions are given in Section V.

Notations. $I_n \in \mathbb{R}^{n \times n}$ is a unit matrix. $\sigma(A)$ denotes the complex spectrum of matrix A and $\otimes$ indicates the Kronecker product operator. $\mathbb{Z}_+$ denotes the set of positive natural numbers. For a vector $a \in \mathbb{R}^n$, define $\text{vecv}(a) = [a_1^2, a_1 a_2, \dots, a_1 a_n, a_2^2, a_2 a_3, \dots, a_n^2]^T \in \mathbb{R}^{\frac{n(n+1)}{2}}$. For a matrix $C \in \mathbb{R}^{m \times n}$, define $\text{vec}(C) = [c_1^T, c_2^T, \dots, c_n^T]^T \in \mathbb{R}^{1 \times mn}$ with $c_i \in \mathbb{R}^m$ the columns of C . For a symmetric matrix $D \in \mathbb{R}^{n \times n}$, define $\text{vecs}(D) = [d_{11}, 2d_{12}, \dots, d_{22}, 2d_{23}, \dots, 2d_{(n-1)n}, d_{nn}] \in \mathbb{R}^{1 \times \frac{n(n+1)}{2}}$.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Preliminaries

Consider a continuous-time linear system given by

$$\dot{x} = Ax + Bu + Dv \quad (1)$$

$$\dot{v} = E_v v \quad (2)$$

$$e = Cx + Fv \quad (3)$$

where $x \in \mathbb{R}^n$, $u \in \mathbb{R}^m$, $v \in \mathbb{R}^q$, and $e \in \mathbb{R}^r$ represent the system state, control input, state of an exogenous system (exosystem), and output error, respectively, and $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{r \times n}$, $D \in \mathbb{R}^{n \times q}$, $E_v \in \mathbb{R}^{q \times q}$, and $F \in \mathbb{R}^{r \times q}$ are constant matrices. In the linear output regulation problem (LORP), $d = Dv$ and $y_d = -Fv$ are generated by the exosystem, representing the reference signal and the disturbance. $y = Cx$ stands for the output. Throughout this paper, some standard assumptions are made as follows.

Assumption 1: The exogenous signal v is unmeasurable.

Assumption 2: The pair (A, B) is stabilizable.

Assumption 3: $\text{rank}\left(\begin{bmatrix} A - \lambda I_n & B \\ C & 0 \end{bmatrix}\right) = n + r, \forall \lambda \in \sigma(E_v)$.

Assumption 4: The minimal polynomial of E_v is available with degree $q_m \leq q$.

Under Assumption 4, we can find a vector $w \in \mathbb{R}^{q_m}$ and a known matrix $E_w \in \mathbb{R}^{q_m \times q_m}$ such that

$$\dot{w} = E_w w \quad (4)$$

$$v = E_{v \leftarrow w} w, \forall t \geq 0 \quad (5)$$

where $E_{v \leftarrow w} \in \mathbb{R}^{q \times q_m}$ is an unknown constant matrix.

Therefore, we can rewrite (1) and (3) as

$$\dot{x} = Ax + Bu + D_w w \quad (6)$$

$$e = Cx + F_w w \quad (7)$$

where $D_w = DE_{v \leftarrow w}$ and $F_w = FE_{v \leftarrow w}$.

In this paper, the objective is to solve the LOORP with the convergence rate of the tracking error requirement, that is, to design an optimal controller u for the system (6) and (7) such that the closed-loop system is asymptotically stable and the predefined cost is minimized. Moreover, the tracking error

converges at a rate faster than that of $e^{-\gamma t}$. As a result, the following requirement should be met

$$\lim_{t \rightarrow \infty} e^{\gamma t} e(t) = 0 \quad (8)$$

with the convergence rate criterion γ satisfying $0 \leq \gamma < \gamma^*$.

Remark 1: Assumptions 2–3 guarantee the solvability of the classic LORP [3] and Assumption 4 requires that the frequency of exogenous signal v be known, which is a standard assumption in classical output regulation problem by internal model principle [10]. γ^* is the maximum value of the convergence rate γ , which will be introduced later.

B. Problem Formulation

Now, we introduce an internal model to solve the LOORP, and the internal model is given as

$$\dot{z} = G_1 z + G_2 e \quad (9)$$

where $z \in \mathbb{R}^{r q_m}$ is the internal model state, and the characteristic polynomial of $G_1 \in \mathbb{R}^{r q_m \times r q_m}$ is equivalent to the minimal polynomial of E_w , and the pair (G_1, G_2) is controllable. Essentially, (G_1, G_2) incorporates an internal model of the matrix E_w . Let $x_z = [x^T, z^T]^T \in \mathbb{R}^{n_z}$ with $n_z = n + r q_m$, we can rewrite (6), (7) and (9) in compact form as

$$\dot{x}_z = A_z x_z + B_z u + D_z w \quad (10)$$

$$e = C_z x_z + F_z w \quad (11)$$

where $A_z = \begin{bmatrix} A & 0 \\ G_2 C & G_1 \end{bmatrix}$, $B_z = \begin{bmatrix} B \\ 0 \end{bmatrix}$, $D_z = \begin{bmatrix} D_w \\ G_2 F_w \end{bmatrix}$, $C_z = [C \ 0]$, and $F_z = F_w$.

By introducing the internal model (9), we present a state-feedback controller as

$$u = -K_x x - K_z z = -K x_z \quad (12)$$

where $K_x \in \mathbb{R}^{m \times n}$, $K_z \in \mathbb{R}^{m \times r q_m}$, and $K = [K_x, K_z]$ is the feedback gain and K should be designed to ensure that $A_z - B_z K$ is Hurwitz.

As discussed in [3], under Assumptions 2–4, for any given pair (G_1, G_2) incorporating the internal model of E_w , if the LORP of systems (6)–(7) is solvable, then there exists a unique solution (X^*, Z^*) that solves the following regulator equations

$$\begin{bmatrix} X \\ Z \end{bmatrix} E_w = \begin{bmatrix} A - B K_x & -B K_z \\ 0 & G_1 \end{bmatrix} \begin{bmatrix} X \\ Z \end{bmatrix} + \begin{bmatrix} D_w \\ 0 \end{bmatrix} \quad (13)$$

$$0 = C X + F_z. \quad (14)$$

Define $\tilde{x} = e^{\gamma t}(x - X^* w)$, $\tilde{z} = e^{\gamma t}(z - Z^* w)$, $\tilde{u} = e^{\gamma t}[u - (-K_x X^* - K_z Z^*) w]$, $\tilde{e} = e^{\gamma t} e$, and $\tilde{x}_z = [\tilde{x}^T, \tilde{z}^T]^T$ as the new system state, internal model state, control input,

tacking error and augmented state, respectively. With the solutions to (13) and (14), we obtain

$$\begin{aligned}\dot{\tilde{x}} &= \gamma e^{\gamma t}(x - X^*w) + e^{\gamma t}(Ax + Bu + D_w w - X^*E_w w) \\ &= \gamma \tilde{x} + e^{\gamma t}[A(x - X^*w) + B(u + (K_x X^* + K_z Z^*)w)] \\ &= \tilde{A}\tilde{x} + B\tilde{u}\end{aligned}\quad (15)$$

$$\begin{aligned}\dot{\tilde{z}} &= \gamma e^{\gamma t}(z - Z^*w) + e^{\gamma t}(G_1 z + G_2 e - Z^*E_w w) \\ &= \gamma \tilde{z} + e^{\gamma t}[G_1(z - Z^*w) + G_2 C(x - X^*w)] \\ &= \tilde{G}_1 \tilde{z} + G_2 C \tilde{x}\end{aligned}\quad (16)$$

where $\tilde{A} = (\gamma I + A)$ and $\tilde{G}_1 = (\gamma I + G_1)$.

Thus, we can get a new continuous-time system as

$$\dot{\tilde{x}}_z = \tilde{A}_z \tilde{x}_z + B_z \tilde{u} \quad (17)$$

$$\bar{e} = C_z \tilde{x}_z \quad (18)$$

where $\tilde{A}_z = A_z + \gamma I$. It is worth noting that if $\lim_{t \rightarrow \infty} x - X^*w = 0$ and $\lim_{t \rightarrow \infty} z - Z^*w = 0$ hold, then $\lim_{t \rightarrow \infty} e = 0$. By integrating (12) with the definitions of $\tilde{x}$, $\tilde{u}$, and $\tilde{z}$, we obtain

$$\tilde{u} = -K_x \tilde{x} - K_z \tilde{z} = -K \tilde{x}_z \quad (19)$$

We consider the LOORP with requirements of asymptotic tracking of the output and an assured convergence rate for the linear system, which implies the following dynamic optimization Problem 1 is required to be solved.

Problem 1:

$$\begin{aligned}\min_{\tilde{u}} \quad & \int_0^\infty (\tilde{x}_z^T Q \tilde{x}_z + \tilde{u}^T R \tilde{u}) dt \\ \text{s.t.} \quad & \dot{\tilde{x}}_z = \tilde{A}_z \tilde{x}_z + B_z \tilde{u}\end{aligned}\quad (20)$$

where $Q = Q^T \geq 0 \in \mathbb{R}^{n_z \times n_z}$, $R = R^T > 0 \in \mathbb{R}^{m \times m}$.

Problem 1 is a standard linear quadratic regulator problem and the optimal state feedback controller is

$$\tilde{u} = -K^* \tilde{x}_z \quad (21)$$

where $K^* = R^{-1} B_z^T P^*$ is the optimal feedback control gain, and the matrix $P^* = (P^*)^T > 0$ is the solution to the following ARE

$$\tilde{A}_z^T P + P \tilde{A}_z + Q - P B_z R^{-1} B_z^T P = 0 \quad (22)$$

Observe that the control gain K in (21) for system (17) is the same as in (12) for system (10). Thus, we can obtain the optimal controller u as

$$u = -K^* x_z, \quad (23)$$

which solves the LOORP while meeting an assured convergence rate requirement.

III. DATA-DRIVEN OPTIMAL CONTROLLER DESIGN

In this section, a virtual system is constructed by some filters and then a data-driven learning algorithm is proposed to solve the LOORP with an assured convergence rate. The algorithm is capable of estimating the optimal control gain K^* by collecting the virtual system data. In particular, the system matrices A , B , C , D , and F are not required for the implementation of the proposed algorithm, while the only knowledge of E_v is its minimal polynomial.

A. Construction of virtual system

In order to meet the need for the convergence rate of the tracking error, define $\bar{x} = e^{\gamma t} x$, $\bar{z} = e^{\gamma t} z$, $\bar{w} = e^{\gamma t} w$, $\bar{u} = e^{\gamma t} u$, and $\bar{x}_z = [\bar{x}^T, \bar{z}^T]^T$, and we have

$$\dot{\bar{x}}_z = \tilde{A}_z \bar{x}_z + B_z \bar{u} + D_z \bar{w}. \quad (24)$$

To relax requirement on data calculation in learning process, motivated by [22], we introduce the following filters to construct the virtual system I

$$\dot{h}_f(t) = -k h_f(t) + \dot{\bar{x}}(t), h_f(0) = 0 \quad (25)$$

$$\dot{x}_f(t) = -k x_f(t) + \bar{x}(t), x_f(0) = 0 \quad (26)$$

$$\dot{u}_f(t) = -k u_f(t) + \bar{u}(t), u_f(0) = 0 \quad (27)$$

$$\dot{w}_f(t) = -k w_f(t) + \bar{w}(t), w_f(0) = 0 \quad (28)$$

$$\dot{e}_f(t) = -k e_f(t) + \bar{e}(t), e_f(0) = 0 \quad (29)$$

where $k \in \mathbb{R}$ and $k > 0$ is a tunable gain, and h_f , x_f , u_f , w_f , and e_f are the results of filtering state $\dot{\bar{x}}$, $\bar{x}$, $\bar{u}$, $\bar{w}$ and $\bar{e}$, respectively.

In view of the unavailable term $\dot{\bar{x}}$, h_f cannot be directly calculated by (25). However, based on the solutions to (25) and (26)

$$h_f(t) = e^{-kt} \int_0^t e^{k\tau} \dot{\bar{x}}(\tau) d\tau, x_f(t) = e^{-kt} \int_0^t e^{k\tau} \bar{x}(\tau) d\tau.$$

By using integration by parts, h_f can be explicitly computed as follows

$$h_f(t) = \bar{x}(t) - e^{-kt} \bar{x}(0) - k x_f(t). \quad (30)$$

Define $\delta = h_f - (\tilde{A} x_f + B u_f + D_w w_f)$. It follows from the filter equations (25)–(29) that $\dot{\delta}(t) = -k \delta(t)$, $\forall t \geq 0$. Since $k > 0$, the virtual system is stable. Thus, as $t \rightarrow \infty$, we rewrite the filters (26)–(29) in compact form as

$$h_f = \tilde{A} x_f + B u_f + D_w w_f. \quad (31)$$

Then, we introduce the virtual system II as follows

$$h_f = \tilde{G}_1 z_f + G_2 e_f \quad (32)$$

where z_f is a bounded signal defined by the user rather than the real internal model signal, and it need to satisfy the rank condition in next subsection.

Define $h_l = [h_f^T, h_f^T]^T$, $x_l = [x_f^T, z_f^T]^T$, $u_l = u_f$, and $w_l = w_f$. Based on (31) and (32), a new augmented virtual system can be formulated as

$$h_l = \tilde{A}_z x_l + B_z u_l + D_z w_l. \quad (33)$$

Note that the system matrices of the virtual system (33) are the same as those of (24) and h_l can be obtained directly from (26) and (30). The properties allow us to design the iterative learning equation without requiring FWIs by virtual system data, which will be introduced in the next subsection.

Algorithm 1 Model-based VI for the LOORP with an Assured Convergence Rate

Initialize. Choose $P_0 = P_0^T > 0$, an arbitrary initial control gain K_0 , a small threshold ε , and a desired convergence rate parameter γ . Select a proper bounded sets $\{\mathcal{B}_i\}_{i=0}^\infty$ and a sequence $\{\varepsilon_j\}_{j=0}^\infty$ satisfying

$$\mathcal{B}_i \subseteq \mathcal{B}_{i+1}, \quad i \in \mathbb{Z}_+, \quad \lim_{i \rightarrow \infty} \mathcal{B}_i = \mathcal{P}_+^n \quad (34)$$

$$\varepsilon_j > 0, \quad \sum_{j=0}^\infty \varepsilon_j = \infty, \quad \sum_{j=0}^\infty \varepsilon_j^2 < \infty \quad (35)$$

loop

$$\tilde{P}_{j+1} \leftarrow P_j + \varepsilon_j (\tilde{A}_z^T P_j + P_j \tilde{A}_z + Q - K_j^T R K_j) \quad (36)$$

$$K_{j+1} \leftarrow R^{-1} B_z^T \tilde{P}_{j+1} \quad (37)$$

if $\tilde{P}_{j+1} \notin \mathcal{B}_i$ **then**

$$P_{j+1} \leftarrow P_0, \quad i \leftarrow i + 1, \quad j \leftarrow j + 1$$

else if $\|\tilde{P}_{j+1} - P_j\| / \varepsilon_j < \varepsilon$ **then**

$$P^* \leftarrow P_j, \quad K^* \leftarrow K_j, \quad \textbf{break}$$

else

$$P_{j+1} \leftarrow \tilde{P}_{j+1}, \quad j \leftarrow j + 1$$

end if

end loop

B. Data-driven optimal control algorithm development

Even if the system model can be obtained, solving the ARE still poses difficulties due to its nonlinear nature. Therefore, some iteration-based approaches have been developed to overcome this difficulty. The following Algorithm 1 shows the way to solve the LOORP by model-based VI algorithm.

The convergence of Algorithm 1 has been shown in [21]. The convergence rate requirement is achievable when the pair $(\tilde{A}_z, B_z)$ is stabilizable. Thus, choose the convergence rate $\gamma < \gamma^*$, where γ^* is the opposite number of the maximum real part of uncontrollable stable eigenvalues of (A_z, B_z) , to ensure that $(\tilde{A}_z, B_z)$ is stabilizable.

Now, we are ready to propose a data-driven learning algorithm to solve Problem 1. Define $T_j = h_l^T P_j x_l + x_l^T P_j h_l$. Substituting the augmented virtual system (33) into T_j yields

$$x_l^T H_j x_l + 2u_l^T R K_j x_l + 2w_l^T D_j x_l = 2h_l^T P_j x_l \quad (38)$$

where $H_j = \tilde{A}_z^T P_j + P_j \tilde{A}_z$, $K_j = R^{-1} B_z^T P_j$, and $D_j = D_z^T P_j$. Furthermore, for a positive integer s , define matrices

$$\begin{aligned} \Gamma_{x_l x_l} &= [\text{vecv}(x_l(t_1)), \text{vecv}(x_l(t_2)), \dots, \text{vecv}(x_l(t_s))]^T \\ \Gamma_{x_l h_l} &= [x_l(t_1) \otimes h_l(t_1), x_l(t_2) \otimes h_l(t_2), \dots, x_l(t_s) \otimes h_l(t_s)]^T \\ \Gamma_{h_l x_l} &= [h_l(t_1) \otimes x_l(t_1), h_l(t_2) \otimes x_l(t_2), \dots, h_l(t_s) \otimes x_l(t_s)]^T \\ \Gamma_{x_l u_l} &= [x_l(t_1) \otimes R u_l(t_1), x_l(t_2) \otimes R u_l(t_2), \dots, x_l(t_s) \otimes R u_l(t_s)]^T \\ \Gamma_{x_l w_l} &= [x_l(t_1) \otimes w_l(t_1), x_l(t_2) \otimes w_l(t_2), \dots, x_l(t_s) \otimes w_l(t_s)]^T \end{aligned}$$

where $t_1 < t_2 < \dots < t_s$ are positive integers.

With the above matrices, define $\psi = [\Gamma_{x_l x_l}, 2\Gamma_{x_l u_l}, 2\Gamma_{x_l w_l}]$ and $\phi_j = [\text{vecs}(H_j)^T, \text{vecs}(K_j)^T, \text{vecs}(D_j)^T]^T$, and (38) implies the following matrix equation

$$\psi \phi_j = (\Gamma_{x_l h_l} + \Gamma_{h_l x_l}) \text{vecs}(P_j). \quad (39)$$

If the following rank condition holds,

$$\text{rank}(\psi) = \frac{1}{2}(n + r q_m)(n + 2q_m + 2m + r q_m + 1) \quad (40)$$

then, (39) is uniquely solved, i.e.

$$\phi_j = (\psi^T \psi)^{-1} \psi^T (\Gamma_{x_l h_l} + \Gamma_{h_l x_l}) \text{vecs}(P_j). \quad (41)$$

Remark 2: In the previous works [9], [10], [13], [15], since the data $\dot{x}$ is unavailable, the authors constructed the ILEs by mixing a number of FWIs with the Kronecker product. However, we can directly obtain the data h_f to construct the ILE (38) without the integral operations through the virtual system (33), which takes advantages in reducing computational complexity. ■

Finally, a novel data-driven learning algorithm is proposed to solve the LOORP, which is shown as Algorithm 2.

Algorithm 2 Data-Driven Learning Algorithm for the LOORP with an Assured Convergence Rate

Initialize. Choose $P_0 = P_0^T > 0$, an arbitrary initial control gain K_0 , a desired convergence rate parameter $\gamma > 0$, and a small threshold ε . Select a proper bounded set $\{\mathcal{B}_i\}_{i=0}^\infty$ and a sequence $\{\varepsilon_j\}_{j=0}^\infty$ satisfying (34) and (35).

Data Collection. Apply $u = -K_0 x_z + \zeta$ as the input and collect data (h_l, x_l, u_l, w_l) from the virtual system (33) for $[t_0, t_s]$, where ζ is the exploration signal, $t_s = t_0 + sT$ and T is the interval length.

loop

Solve for matrices H_j and K_j by (41), and update $\tilde{P}_{j+1}$ as

$$\tilde{P}_{j+1} \leftarrow P_j + \varepsilon_j (H_j - K_j^T R K_j + Q) \quad (42)$$

if $\tilde{P}_{j+1} \notin \mathcal{B}_i$ **then**

$$P_{j+1} \leftarrow P_0, \quad i \leftarrow i + 1, \quad j \leftarrow j + 1$$

else if $\|\tilde{P}_{j+1} - P_j\| / \varepsilon_j < \varepsilon$ **then**

$$P^* \leftarrow P_j, \quad K^* \leftarrow K_j, \quad \textbf{break}$$

else

$$P_{j+1} \leftarrow \tilde{P}_{j+1}, \quad j \leftarrow j + 1$$

end if

end loop

It should be noted that ζ is an exploration noise introduced when we collect the system data to estimate the unknown optimal solution. Specifically, ζ is necessary to ensure that the rank condition (40) is satisfied.

Remark 3: (39) is uniquely solved while the rank condition (40) is satisfied. This implies Algorithm 2 is equivalent to Algorithm 1. Thus, we have $\lim_{j \rightarrow \infty} P_j = P^*$, $\lim_{j \rightarrow \infty} K_j = K^*$ by completing Algorithm 2. The major difference from [21] is

that our approach includes a user-defined internal model, with data from the virtual system (33). ■

Now, the main result is summarized as the following theorem.

Theorem 1: Consider a continuous-time linear system (10)–(11). Under Assumptions 1–4 and an appropriate convergence rate γ , the controller $u = -K_j x_z$ obtained by Algorithm 2 makes the closed-loop system asymptotically stable. Moreover, it guarantees that the state and the tracking error converge faster than $e^{-\gamma t}$. ■

Proof 1: The system (10)–(11) with the estimated optimal controller $u = -K_j x_z$ by Algorithm 2 has an equivalence with the following system

$$\dot{\tilde{x}}_z = (\tilde{A}_z - B_z K_j) \tilde{x}_z \quad (43)$$

$$\bar{e} = C_z \tilde{x}_z. \quad (44)$$

The control gain matrix K_j estimate by Algorithm 2 converges to K^* as $j \rightarrow \infty$. Thus, $(\tilde{A}_z - B_z K_j)$ is Hurwitz, $\lim_{t \rightarrow \infty} \tilde{x}_z(t) = 0$ holds. Then, we have

$$\lim_{t \rightarrow \infty} \bar{e}(t) = \lim_{t \rightarrow \infty} C_z \tilde{x}_z(t) = 0. \quad (45)$$

Based on the definitions of $\tilde{x}$ and $\bar{e}$, $x(t)$ and $e(t)$ converge faster than $e^{-\gamma t}$. This completes the proof. ■

IV. ILLUSTRATIVE EXAMPLE

To show the effectiveness of the our approach, consider the following LCL coupled inverter-based distributed generation system as in [10], [15],

$$\begin{bmatrix} \dot{I}_L \\ \dot{V}_C \\ \dot{I}_O \end{bmatrix} = \begin{bmatrix} -\frac{R_1}{L_1} & -\frac{1}{L_1} & 0 \\ \frac{1}{C} & 0 & \frac{1}{C} \\ 0 & \frac{1}{L_2} & -\frac{R_2}{L_2} \end{bmatrix} \begin{bmatrix} I_L \\ V_C \\ I_O \end{bmatrix} + \begin{bmatrix} \frac{1}{L_1} \\ 0 \\ 0 \end{bmatrix} V_I + \begin{bmatrix} 0 \\ 0 \\ -\frac{V_G}{L_2} \end{bmatrix}$$

where I_L , I_O , V_C , V_I and V_G denote the inductor current, the output current, the capacitor voltage, the input voltage, and the grid voltage, respectively. The the output is $y = I_O$, the reference signal is $y_d(t) = -10 \sin(100\pi t + \pi/6)$, and the tracking error is $e = y - y_d$.

In contrast to the standard feedforward-feedback based approach in [10], [15], which requires partial system matrices, we solve the LOORP when the system matrices are completely unknown by employing internal model principle. We choose

$$G_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -4 \times 10^8 \pi^2 & 0 & -5 \times 10^4 \pi^2 & 0 \end{bmatrix}, G_2 = [0, 0, 0, 1]^T$$

and set $w(0) = [0, 1, 0, 4]^T$. We set the tuning gain $k = 500$, the convergence rate parameter $\gamma = 0.3$, and $Q = I_7$ and $R = 1$. To show the effectiveness of Algorithm 2, we first calculate the K^* via Algorithm 1 with the system model and K^* is obtained as

$$K^* = \begin{bmatrix} 15.1966 & 1.2794 & 15.7917 & 0.4018 \\ 26.6454 & -2.5279 & 36.5882 \end{bmatrix}.$$

In this simulation, some important parameters are given as $P_0 = 0.1I_7$, $B_i = \{P \in P^7 : P > 0, \|P\| < 300(i+1)\}$, $\varepsilon_j = 200/(j+10000)$, $\varepsilon = 0.01$, and $\zeta = 10(\sin(5t) + 0.2\sin(8t) + 0.3\sin(10t))$. By collecting the data from the virtual system from $t = 5\text{ms}$ to $t = 10\text{ms}$, we can estimate the optimal control policy gain K_j via the proposed Algorithm 2 and update control policy after $t = 10\text{ms}$. The learned value of K_j is obtained as

$$K_j = \begin{bmatrix} 15.1944 & 1.2791 & 15.7836 & 0.4027 \\ 26.6404 & -2.5231 & 36.5802 \end{bmatrix}.$$

Fig. 1 shows that P_j can converge to the optimal value P^* . The evolutions of the output, reference, and control input are shown in Figs. 2–3. Fig. 2 shows the output of the system can asymptotically track the reference signal. Fig. 4 shows the tracking error converges faster than $e^{-0.3t}$ when the control policy updates. Finally, as shown in Fig. 5, tracking error $e(t)$ can converge at different rates by our approach through the regulation of the convergence rate parameter γ .

V. CONCLUSIONS

This paper develops a data-driven learning algorithm for solving the LOORP with convergence rate requirement. Without the dynamics of the plant system and exo-system, we design a virtual system to replace the original system and the internal model. A novel learning algorithm is developed, which can effectively estimate the optimal controller while reducing the computational load. Future work will consider the transient performance of the system and explore data-driven control algorithms for nonlinear systems.

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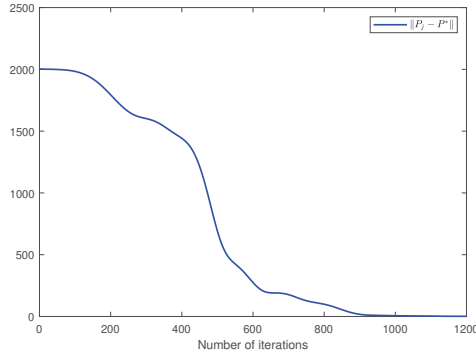


Fig. 1. Evolution of $\|P_j - P^*\|$.

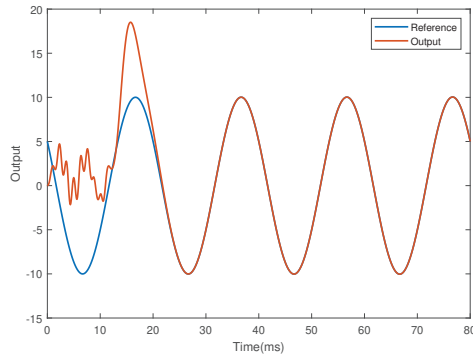


Fig. 2. Evolutions of the output and reference.

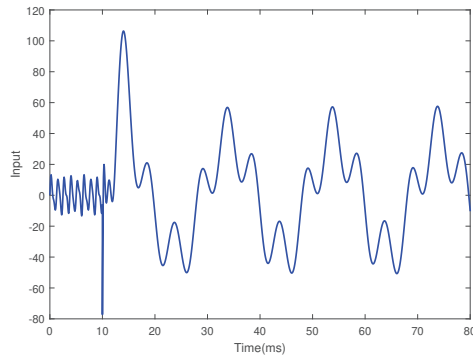


Fig. 3. Evolution of control input $u(t)$.

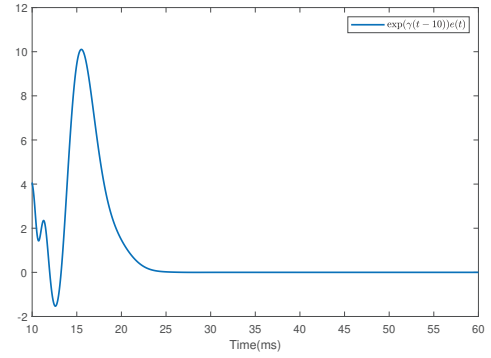


Fig. 4. Trajectory of $e^{\gamma(t-10)}e(t)$.

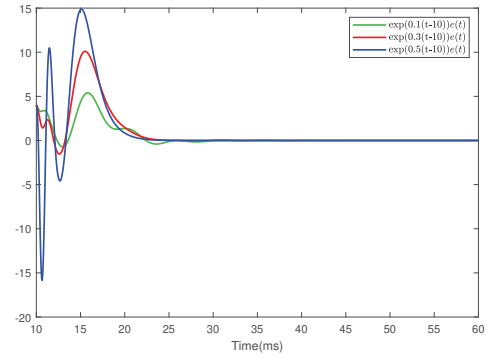


Fig. 5. Simulation results with different convergence rate criterion γ .

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